Machine Learning Based Algorithm Selection for Sorting Numeric Arrays

Official MSc thesis defense

93.1%
runtime gap closed

76.1%
test accuracy

89.6%
zero regret

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The problem is choosing the right algorithm for each input

The thesis question is not how to sort, but which sorter to trust for this array.

Same complexity does not mean same runtime

O(n log n)

same high-level complexity class

introsort

heapsort

timsort

The practical winner depends on input structure, constants, memory behavior

This thesis selects before sorting begins

The selector pays a small feature cost to avoid a larger runtime mistake.

The aim is a practical selector, not a perfect oracle

93.1%
gap closed

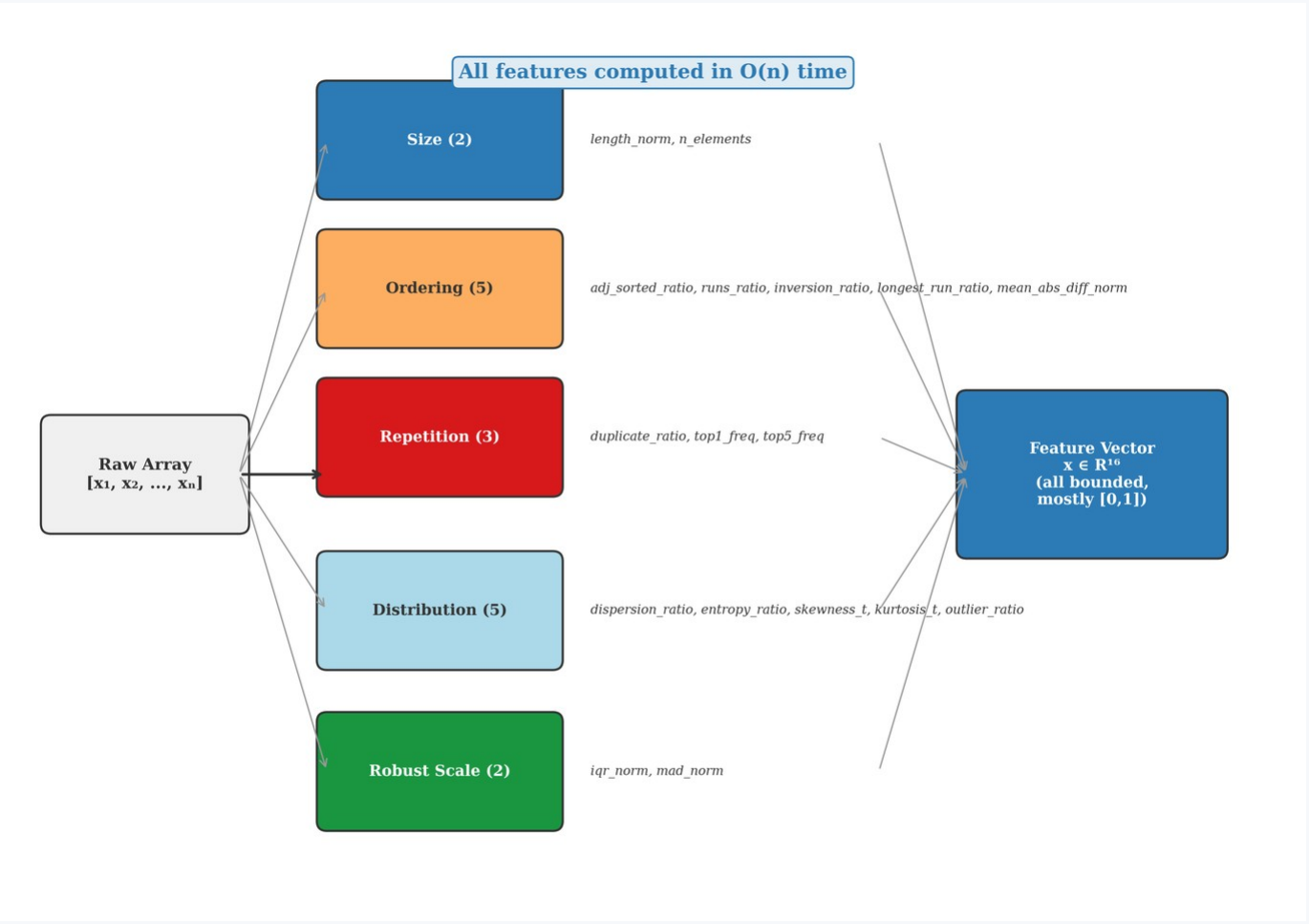
76.1%
accuracy

89.6%
zero regret

A wrong label is only damaging when it creates meaningful runtime regret.

That is why the defense leads with runtime value, then explains accuracy and limits.

The array is converted into structural signals

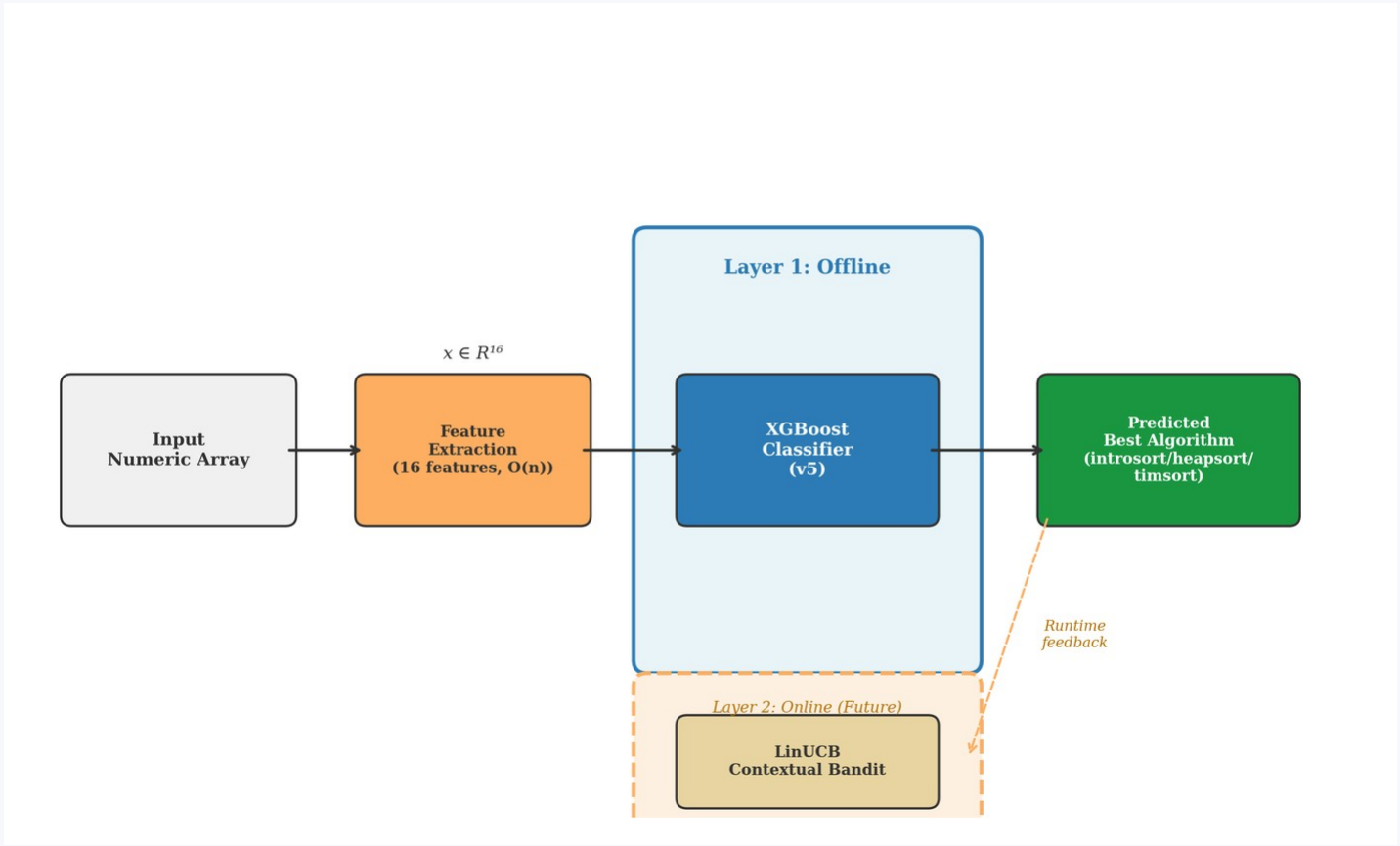


The model does not see the raw sequence.

It sees compact structure:

size
ordering
repetition
distribution

The system makes one decision before execution



features

XGBoost

one of three algorithms

No multiple sorting runs are used at de

The algorithms are different enough to make selection useful

introsort

hybrid quicksort path

strong average behavior

heapsort

stable worst-case bound

often the SBS baseline

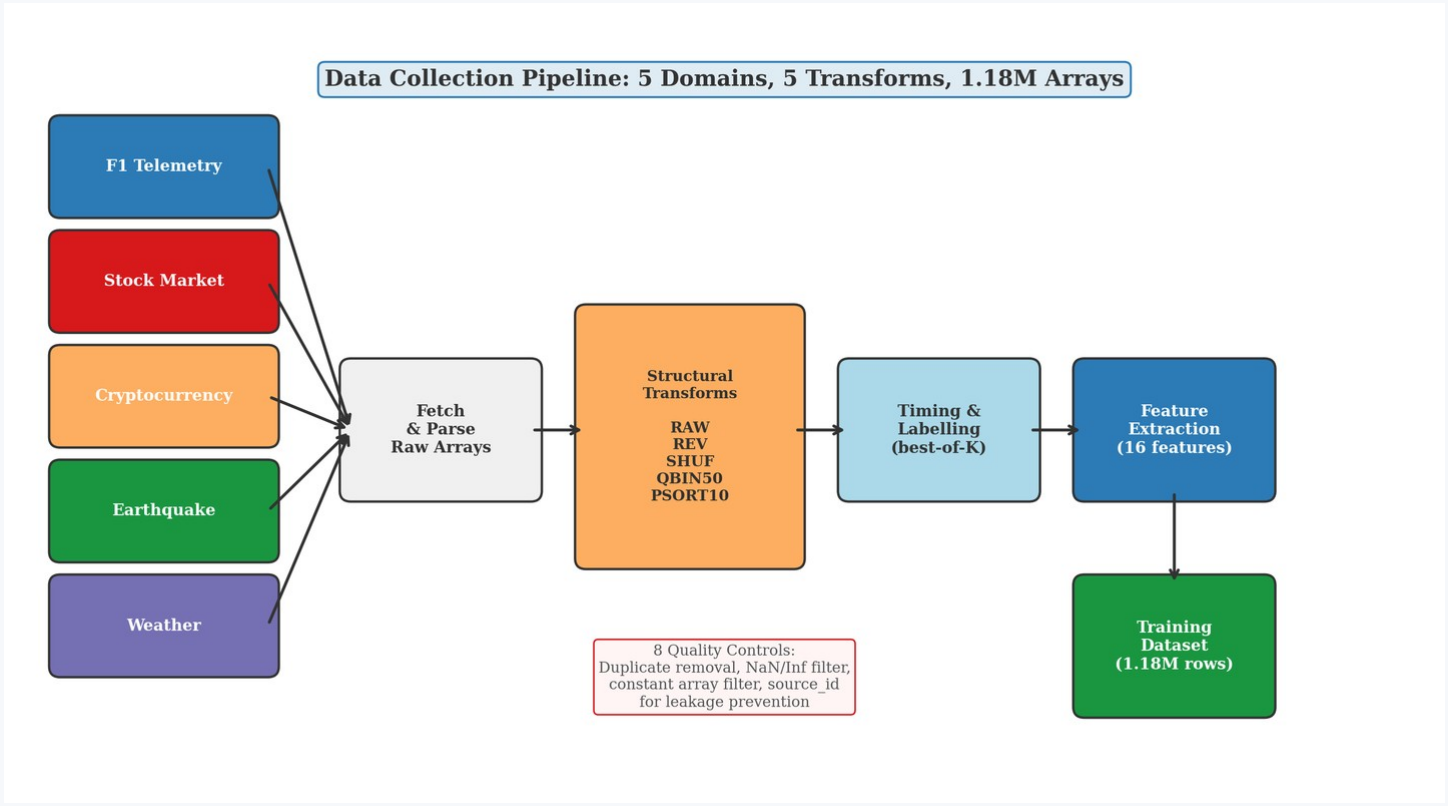
timsort

uses existing runs

structurally visible

The portfolio is small enough to defend and different enough to learn.

Real arrays give the selector the structures synthetic data missed



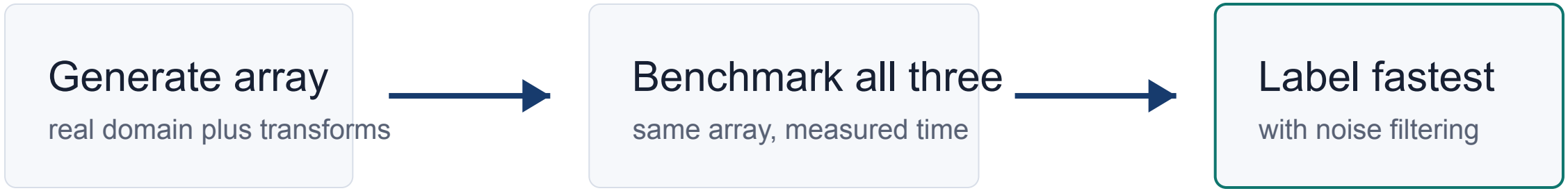
1.18M
arrays before filtering

5 domains

F1 telemetry
stock, crypto
earthquake, weather

Source: metrics_summary.json M5_dataset_statistics

The labels come from measured runtime, not opinion



Transforms used: **RAW / REV / SHUF / QBIN50 / PSORT10**

The model learns from observed winners, not from a manual rule about algorithms.

The features describe structure, not raw values

Size

length_norm

Repetition

duplicate_ratio, top frequency ratios

Ordering

runs, inversions, adjacent sortedness

Distribution

entropy, skewness, dispersion, outliers

16 features summarize what the sorting algorithms can exploit.

The main model is the general v5 selector

XGBoost v5

16 structural features

3 algorithm labels

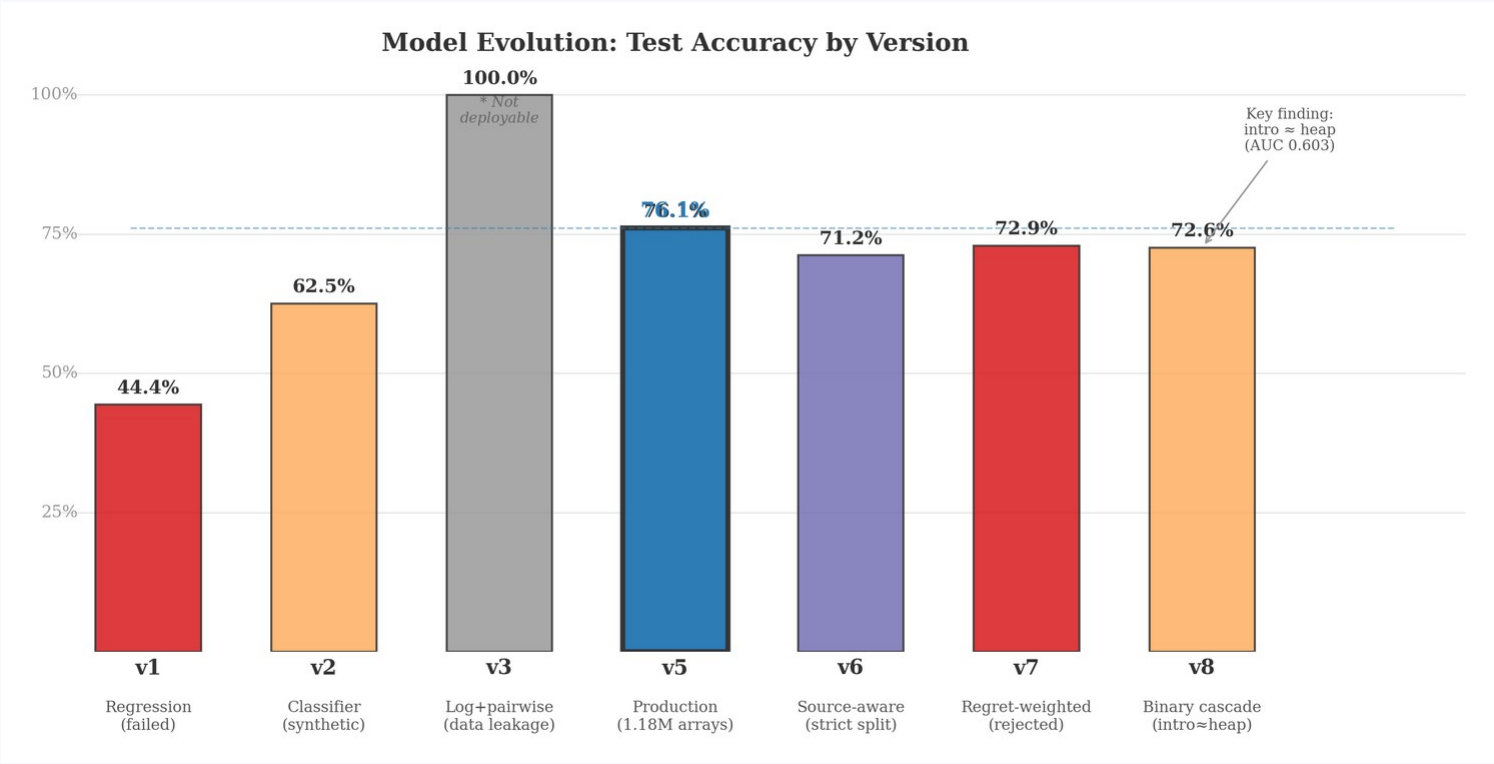
70 / 15 / 15 split

balanced training strategy

Separate from F1 routing

v5 is the general thesis model. F1 channel models are a specialized experiment.

Each failed path clarified the final design



What changed

regression was weak

timing features leaked

regret weighting hurt

cascade exposed boundary

The journey matters because it explains why v5 is the defensible center of the thesis.

The model is imperfect, but most mistakes are cheap

93.1%

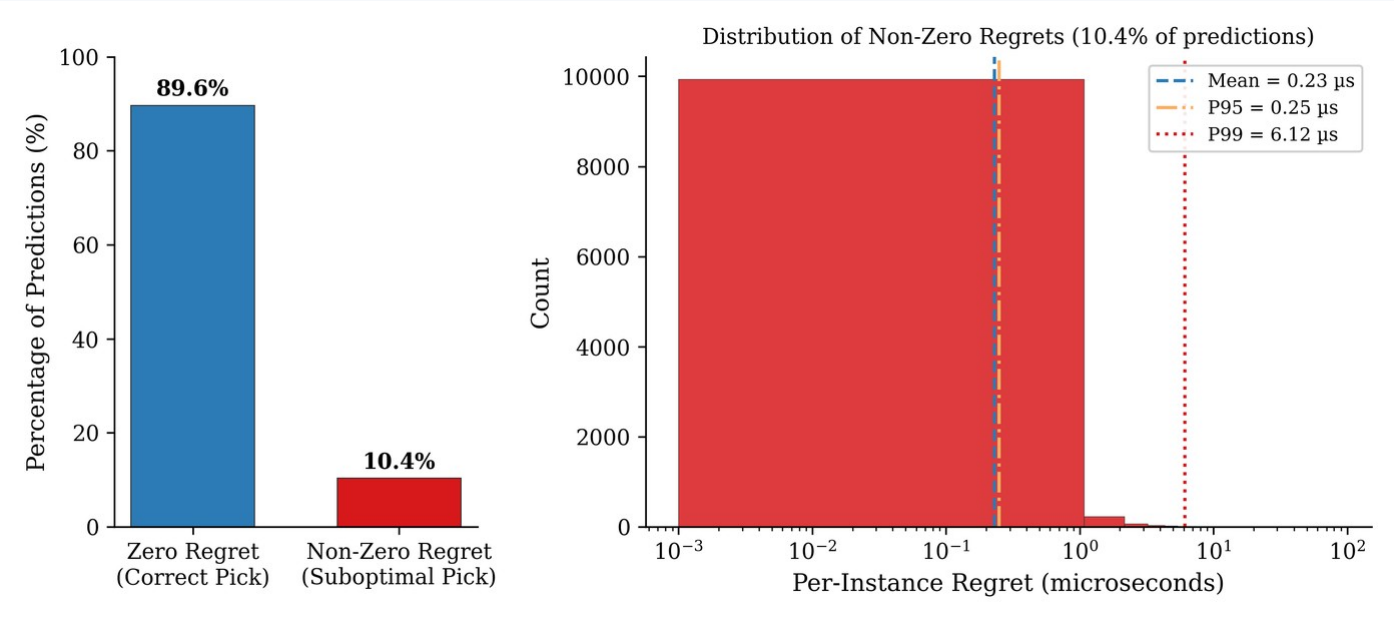
of the SBS-to-VBS runtime gap closed

Supported by 76.1% test accuracy
and 89.6% zero-regret selections.

runtime value first

Source: metrics_summary.json M1_v5_evaluation and M2_v5_regret

Accuracy alone does not explain the value

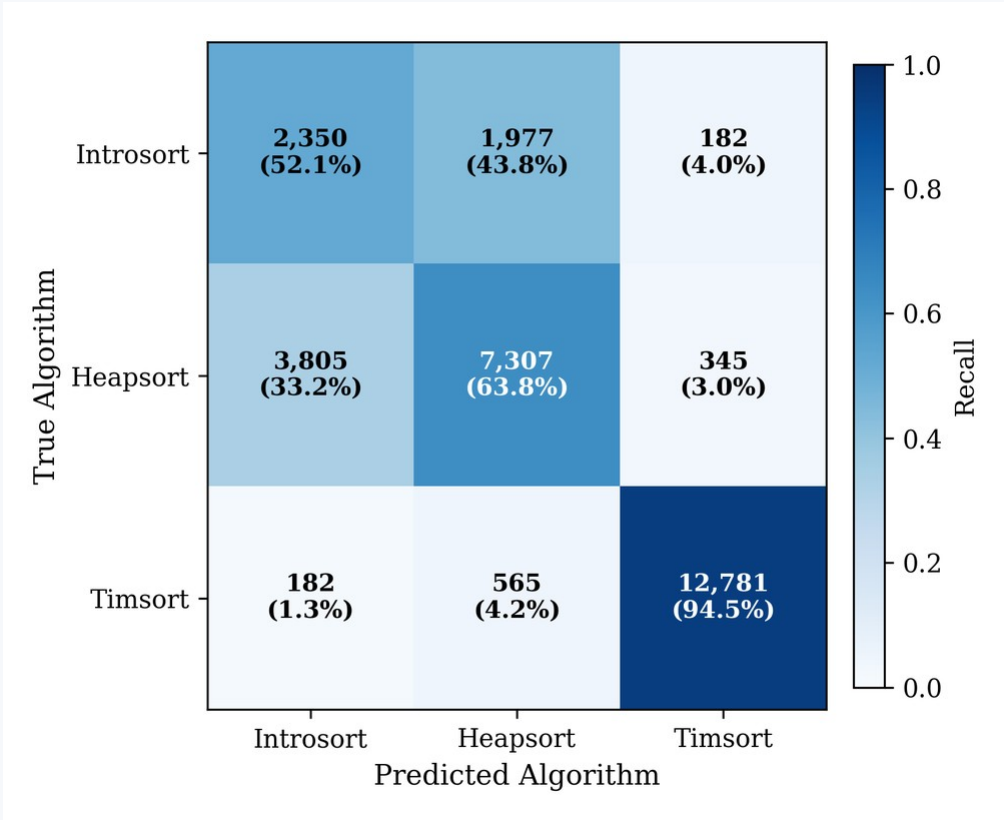


$$Regret = \frac{T_{model} - T_{VBS}}{T_{VBS}}$$

$$Gap\ Closed = \frac{T_{SBS} - T_{model}}{T_{SBS} - T_{VBS}}$$

A selection can be wrong as a label
and still almost right as a runtime decision.

Timsort is structurally visible; introsort vs heapsort is not

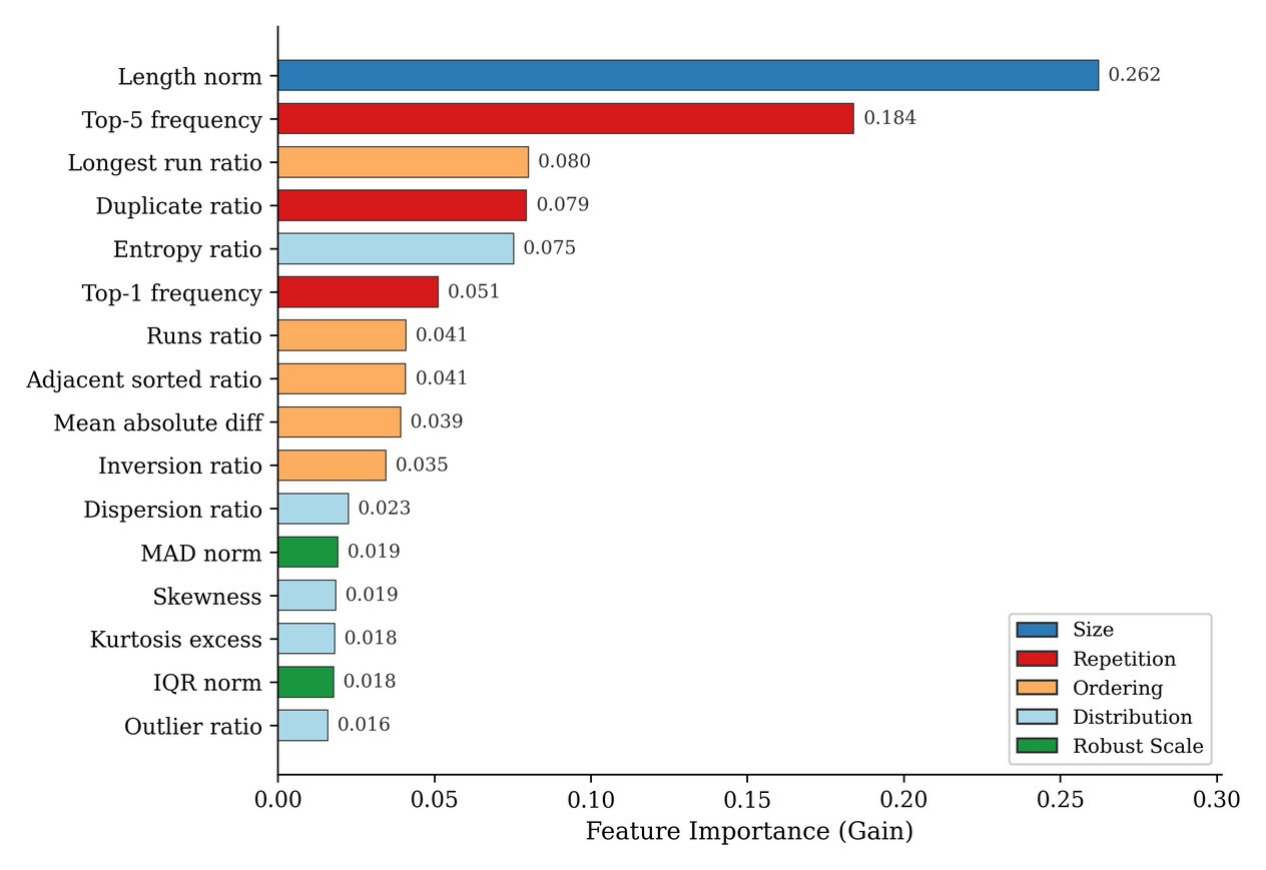


94.5%

timsort recall

The weak boundary is between introsort and heapsort, where structural features are close.

The model mostly uses features that make sorting sense



Top signals

length

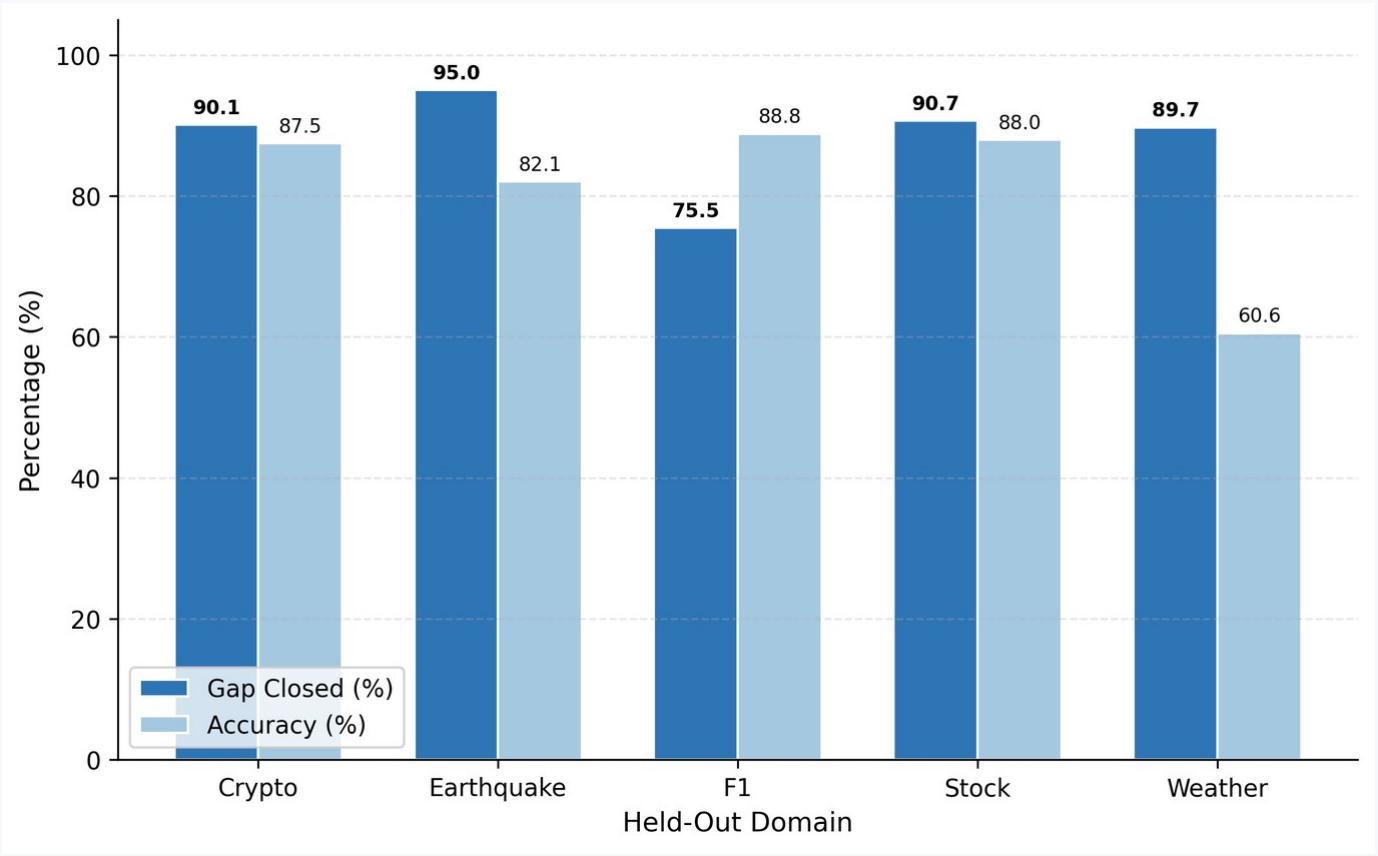
top frequency

longest run

duplicate ratio

This is interpretation, not causal proof.

The selector transfers when structure transfers



Holdout test

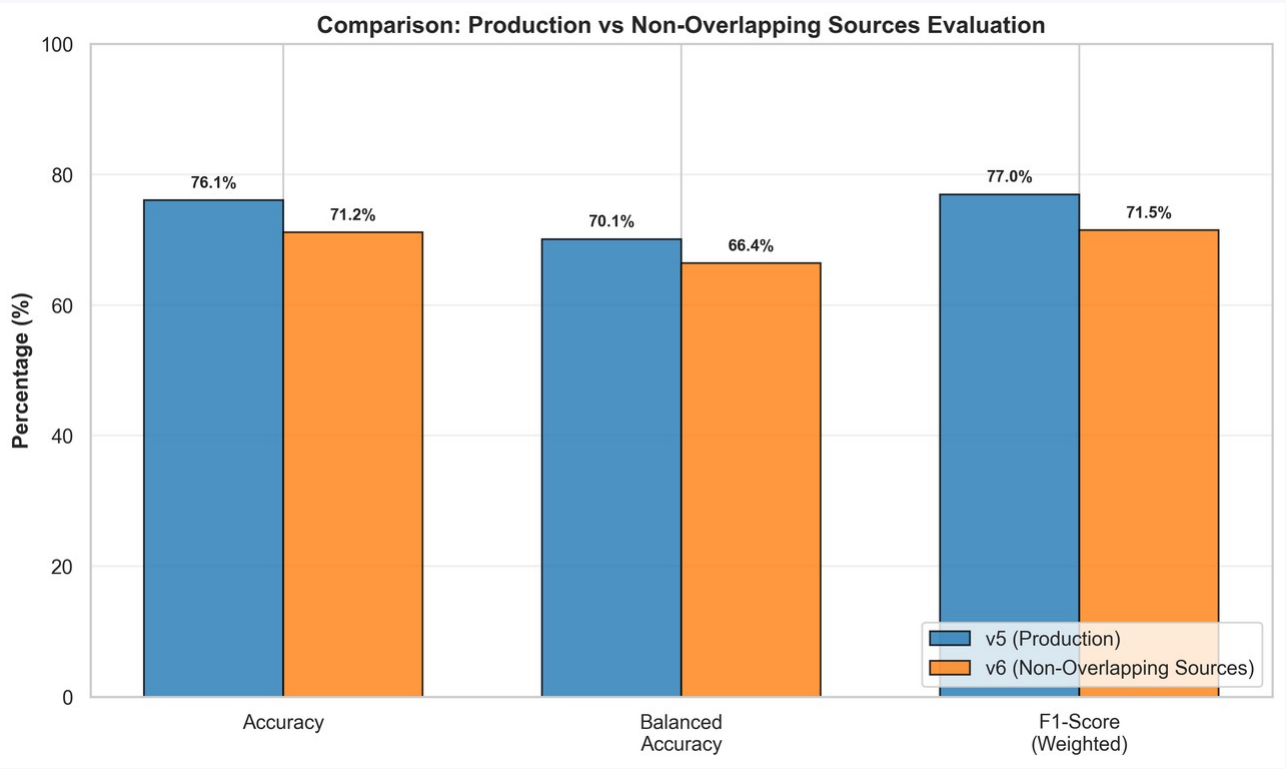
not uniform

but runtime value

survives in several

domain removals

Strict checks changed the interpretation, not the contribution



source-aware split lowered the numbers

v7 regret-aware training was rejected

v8 exposed the intro-heap boundary

The stricter checks make the final claim narrower, but more defensible.

The remaining errors show where the features stop seeing

Clear boundary

timsort vs the rest

Hard boundary

introsort vs heapsort

F1 channel routing is a separate specialization:
useful to mention, not the main thesis model.

This keeps the three-label v5 selector and the F1-specific five-label experiment separated.

The contribution is a selector and an evaluation discipline

1. Dataset and labeling protocol

real domains, structural transforms, measured winners

2. Structural ML selector

v5 XGBoost model using 16 cheap array features

3. Runtime-aware evaluation

accuracy, regret, SBS, VBS, and gap closed together

The next step is hardware-aware and adaptive selection

hardware context

cache, CPU, memory behavior

adaptive selection

future online learning

broader portfolio

more algorithms, stricter splits

?

Thank you

I am ready for your questions.